Comparing the Biomechanics of Professional and Amateur Tennis Serves Using Computer Vision

A Pose-Estimation Case Study of Djokovic, Federer, and an Amateur Player

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DATA 4950 - Research in Data Science
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Spring 2026

Abstract

This project uses computer vision to study the tennis serve through analysis of three serves: two from professional players (Novak Djokovic and Roger Federer) and one from an amateur player. Using MediaPipe Pose, a machine learning-based pose detection system, I identified 33 body landmarks in each video frame and computed joint angles for the elbow, knee, and shoulder throughout the serve motion. I also developed a biomechanics-based scoring function to automatically detect the exact frame of ball impact. The results reveal clear differences between the professionals and the amateur. At impact, both pros achieved near-full elbow extension (approximately 180°), while the amateur reached only 157°-a 23° gap that directly reduces power and consistency. The pros also demonstrated greater knee extension (178.6° and 173.0° versus 163.9°), confirming that professional serves rely on coordinated leg drive. Surprisingly, the amateur's wrist velocity was three times higher than Djokovic's, indicating that amateurs compensate for weak kinetic chain mechanics by over-relying on wrist snap. These findings provide quantitative evidence for the long-established coaching principle of the kinetic chain: power flows from the ground up, not from the wrist.

1. Introduction

Tennis is a sport where small details make enormous differences, and the serve is the clearest example. A great serve can win a point instantly, while a weak one loses it before play begins. For amateur players, the serve is also one of the hardest shots to master because many body parts must move together in precise sequence and timing.

Traditionally, analyzing a serve meant watching slow-motion video or paying a coach to evaluate form. This approach works but is slow, subjective, and depends heavily on the observer's expertise. Today, with open-source pose estimation libraries, Python, and affordable video equipment, it is possible to analyze serves objectively. We can record any player, automatically detect joint positions, and compute the actual angles the body makes during motion.

This project applies that approach. I built a computer vision pipeline that takes a video of a tennis serve, runs MediaPipe Pose on every frame to detect body landmarks, and extracts the joint angles most relevant to the serve: elbow, knee, and shoulder. I then compared these measurements across three players of very different skill levels. The goal is not to revolutionize sports science but to demonstrate that a simple computer vision project can capture the biomechanical differences that coaches discuss informally every day.

2. Problem Statement

The central question is simple to ask but difficult to answer precisely: what measurably separates a professional tennis serve from an amateur one? Everyone agrees that professionals serve better, but "better" is vague. Is it full arm extension? Stronger leg drive? Faster wrist motion? Some combination of these? Without quantitative data, it is hard to know which factors truly matter.

This project breaks that broad question into more concrete, measurable questions:

- How fully does each player extend the elbow at the moment of ball impact?
- How much knee extension (leg drive) do they generate at impact?
- How fast is the wrist moving, and does faster wrist speed indicate a better serve?
- Can accessible computer vision tools reliably detect these differences from standard video?

By converting vague observation into specific measurements, data science makes the question answerable.

3. Methodology

The complete pipeline has three main stages: pose detection from video, joint angle extraction, and automatic impact frame detection. All code is written in Python and available on GitHub (github.com/GonzaloAngelTano/tennis-serve-ai).

3.1 Study Design and Data Selection

This project is structured as a case study analyzing three serves-one each from Novak Djokovic, Roger Federer, and an amateur player. Case studies are appropriate for proof-of-concept work: they validate methodology before scaling. The goal is to demonstrate that computer vision can extract meaningful biomechanical patterns from standard video, not to make statistical claims about all professionals or all amateurs based on limited data.

Initially, I attempted to analyze all serves detected automatically across the three videos using an automated serve-detection algorithm. The algorithm identified 45+ serves total (11 from Djokovic, 32 from Federer, 2 from amateur). However, when validating these detections manually, I found that the scoring function was inconsistent across different video qualities and frame rates, resulting in noisy angle measurements with unexpectedly high standard deviations.

Rather than present unreliable statistics, I made a deliberate methodological choice: select the 3 cleanest, most reliable serves (one from each player) and analyze them with full frame-by-frame validation. This approach prioritizes data quality over sample size-a principle common in biomechanics research where one well-measured motion can be more informative than 50 poorly-detected ones.

The three serves selected showed clear, unobstructed pose detection, consistent angle measurements across frames, well-defined impact moments, and no obvious detection errors when validated manually. This is a trade-off familiar in research: smaller n with high confidence in individual measurements versus larger n with measurement uncertainty. For a proof-of-concept study, quality takes precedence.

All three videos show the player from a side view, the standard angle for biomechanics research because it clearly shows arm and leg motion. The professional clips were downloaded from public sources; the amateur clip was recorded with a standard mobile camera. Video resolution and frame rates varied, but MediaPipe's pose detection normalizes these differences automatically.

3.2 Pose Detection

Pose estimation was performed using Google's MediaPipe Pose library, which detects 33 body landmarks per frame using a deep neural network trained on motion capture data. MediaPipe returns normalized (x, y) coordinates-independent of video resolution-making it practical to compare clips from different sources. I focused on right-side landmarks (shoulder, elbow, wrist, hip, knee, ankle) since all three players served with the right arm.

Each video was processed frame by frame. If MediaPipe could not detect a pose (due to motion blur or obscured joints), that frame was skipped; this is a minor limitation in early or late frames but does not affect the critical impact moment.

3.3 Joint Angle Extraction

After pose detection, I computed three joint angles per frame using the law of cosines. Given three points A, B, and C, the angle at B is:

$$angle(B) = \arccos\left((BA \cdot BC) / (|BA| \cdot |BC|) \right)$$

The three angles tracked were:

- Elbow angle (shoulder → elbow → wrist): measures arm extension
- Knee angle (hip → knee → ankle): measures leg extension
- Shoulder angle (hip → shoulder → elbow): measures serving-arm height

All angle values and metadata were saved to CSV files for later analysis.

3.4 Impact Frame Detection

Detecting the exact frame where the racket contacts the ball is critical but challenging-the ball is usually too small and blurry for object detection. Instead, I used a biomechanics-based approach: at impact, three things happen simultaneously. The arm reaches near-full extension (high elbow angle), the legs push maximally upward (high knee angle), and the wrist reaches its highest point in the frame.

This insight led to a scoring function:

$$score = elbow\_angle + knee\_angle - 300 \times wrist\_height$$

The constant 300 was chosen empirically to balance the angle terms (in degrees, typically 0-180) against wrist height (normalized, 0-1). The frame with the maximum score is marked as impact. Visual inspection confirmed that this method correctly identified the contact frame in all three videos.

3.5 Comparative Analysis

After detecting each player's impact frame, I extracted a window of ±40 frames around it, standardizing the time window for comparison despite different video lengths. Wrist velocity was computed frame-by-frame as the distance between consecutive wrist positions. All curves were lightly smoothed to reduce pose-detection noise. Angular and linear velocities were derived for deeper analysis.

4. Results

The key numerical results are shown in the table below, all measured at the detected impact frame.

Metric	Djokovic	Federer	Amateur
Elbow angle at impact	~180°	~180°	~157°
Knee angle at impact	178.6°	173.0°	163.9°
Max wrist velocity (normalized)	0.0031	---	0.0109

4.1 Elbow Angle at Impact

The elbow angle showed the largest and most consistent difference between professionals and the amateur. Both Djokovic and Federer reached approximately 180° at impact, indicating full arm extension when the racket met the ball. The amateur reached only about 157°, a difference of 23°.

This gap is significant. A fully extended arm puts the racket at maximum height, improving serve trajectory into the court. More importantly, it allows the entire body's rotational energy to transfer efficiently to the ball. When the elbow is bent at impact, much of that energy is lost, and the player must compensate using arm muscles alone-a strategy that is both less powerful and mechanically harder on the joints. The 23° difference is not merely a number; it is the primary reason the amateur's serve is weaker and less reliable.

4.2 Knee Angle at Impact

Knee extension also differed meaningfully between groups, though the absolute difference was smaller. Djokovic reached 178.6°, Federer 173.0°, and the amateur 163.9°. While 10-15° may seem small, in biomechanics it represents a significant portion of available range and directly indicates how much upward force the legs are generating at the critical moment.

The amateur's knees were still substantially bent at impact, suggesting that leg drive either ended prematurely or never fully engaged. Watching the video manually confirmed this observation: the amateur's serve looks like an arm-only motion, without the upward jump or push from the ground that characterizes professional serves.

4.3 Wrist Velocity

The wrist velocity result was the most surprising. Before analysis, I expected professionals to have faster wrist motion, since they serve harder and faster. Instead, the data showed the opposite: the amateur's maximum wrist velocity (0.0109 normalized units) was more than three times higher than Djokovic's (0.0031).

Manual frame-by-frame inspection confirmed the measurement was correct. What is actually happening is that the amateur relies almost entirely on wrist snap to generate power, producing a short, very fast wrist motion. Djokovic, by contrast, has a smooth, controlled wrist because his power comes earlier in the kinetic chain-from legs, trunk, and shoulder. A faster wrist is not a sign of a better serve; it can be a sign of compensation for weak mechanics.

5. Discussion

The three measurements paint a coherent picture. Professionals win on elbow extension, win on knee drive, and paradoxically "lose" on wrist velocity-but this apparent loss is actually a strength. The data reveals the kinetic chain principle in action: great serves generate force starting from the ground (legs push), passing through the trunk (rotation), then shoulder, then arm, and finally wrist (which contributes a small, smooth final motion). The amateur skipped the early links and tried to do everything with the wrist-a mechanically inefficient strategy.

From a data science perspective, the project demonstrates that a relatively simple pipeline can extract meaningful biomechanical insights. I did not use deep learning models trained from scratch, expensive motion-capture suits, or specialized sports-science equipment-just standard video, MediaPipe, and a scoring function derived from domain knowledge. The impact-detection algorithm worked better than expected, showing how domain expertise can make simple algorithms competitive with more complex ones.

5.1 Methodological Rigor and Data Quality

A key decision in this project was prioritizing data quality over sample size. While 45 automatically-detected serves might initially seem preferable to 3 hand-selected ones, the validation process revealed substantial inconsistencies in the automated detection across different video conditions. Rather than present statistics with high measurement uncertainty, I chose to analyze 3 serves that I could validate frame-by-frame for accuracy.

This decision reflects a principle in experimental biomechanics: one carefully-measured observation is more scientifically valuable than 50 noisily-detected ones. The trade-off accepts smaller n in exchange for confidence in the individual measurements. With this approach, I can say with certainty that the measured angles are correct, whereas the larger dataset could not support that claim.

5.2 Limitations

A single serve per player is not fully representative of that player's typical serve-variation can occur between serves, and fatigue or conditions matter. The three videos were not shot under identical conditions (camera angle, lighting, distance). MediaPipe is a 2D system and cannot fully capture motion perpendicular to the camera view, though for serves (mostly vertical) this is acceptable. A production study would require 20-30 serves per player across multiple skill levels, enabling statistical testing and population-level claims.

6. Conclusion

This case study used computer vision to quantify differences between professional and amateur tennis serves. The MediaPipe-based pipeline-landmark detection, angle extraction, and impact detection-produced clear numerical patterns consistent with established biomechanics principles.

Key findings: at impact, the amateur was 23° short of full elbow extension and 15° short of full knee extension compared to the professionals. The amateur also generated wrist velocity three times higher, confirming that power comes from kinetic chain coordination, not wrist speed. These results provide quantitative evidence for what coaches have long taught intuitively.

This project demonstrates that accessible computer vision tools can study real athletic movements meaningfully. It started as curiosity about why my serve does not look like Djokovic's and evolved into a structured analysis with reproducible methodology. Future work would scale to more serves and skill levels, but even in its current form, the project shows the potential of democratized computer vision for sports science.

And practically speaking: I now know exactly what to work on. Straighten that arm.

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